Webinar-01

Logic to Learning: Understanding Gates and Regression

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How Regression "Learns"
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What Regression Really Does

- Regression **models the relationship** between dependent and independent variables
- It explains **how much Y changes when X changes**
- Unlike classification, regression predicts **continuous values**
- *Regression is about trends, not exact predictions*

Weight	height	age
42	160	26
69	195	38
63	151	40
69	187	39
81	182	39
80	175	39
86	195	40
74	191	34
88	181	40
59	155	34
42	164	22

Why Regression Is Important

- Helps in prediction, estimation, and decision-making
- Widely used in economics, management, engineering, health, and ML
- Forms the foundation of many advanced ML algorithms

•Develop an optimized RAG framework for domain-specific applications.
•Create a scalable, efficient, and adaptable system for accurate and contextually relevant responses.

Types of Regression

- Simple Linear Regression → one input, one output
- Multiple Linear Regression → multiple inputs
- Polynomial Regression → captures non-linear patterns
- Logistic Regression → classification (despite the name)

Regression Equation (Core Concept)

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- β_0 → intercept
- β_1 → slope (impact of X on Y)
- ε → error term (unexplained variation)

. Assumptions of Linear Regression

- Linearity
- Independence of errors
- constant variance
- Normality of residuals
- No multicollinearity (for multiple regression)
- *If assumptions fail, results become unreliable.*

How regression Learn

- Minimizes **error between actual and predicted values**
- Uses **Least Squares Method**
- Best-fit line minimizes total squared error

Model Evaluation Metrics

- **MSE / RMSE** → magnitude of error
- **MAE** → average absolute error
- **R^2** → explanatory power of the model
- **Adjusted R^2** → penalizes unnecessary variables
- *High R^2 does not guarantee good predictions.*

Limitations of Regression

- Sensitive to **outliers**
- Assumes linearity (basic models)
- Poor performance with highly complex relationships
- *Regression explains association, not cause*

Regression vs Other ML Models

- Simple and fast
- Works well with small datasets
- Serves as a **baseline model**
- Advanced models are often compared against regression

Thank You